PROJECT MANAGER'S HANDBOOK

Applied Earned Value Management, Commercial Capital Appraisal & Risk Governance

Author: Frank Ellingsen, Financial Controller & Project Control Specialist
Case Study Base: Offshore EPC Platform Drill Tower Construction Project ($BAC = $26.50M)
Target Audience: Project Managers, Project Controllers, CFOs, Steering Committee Members

Handbook Chapter Sitemap

Chapter 1: Principles of Earned Value Management (EVM)

Earned Value Management (EVM) is a systematic project management technique for measuring project performance and progress by integrating Project Scope, Time Schedule, and Actual Cost metrics into a single objective baseline. EVM establishes three foundational metrics at any given status date:

1. Core EVM Triad Definitions:

• **Planned Value (PV):** The authorized budget assigned to scheduled work to be accomplished.

• **Earned Value (EV):** The measure of work performed expressed in terms of the budget authorized for that work.

• **Actual Cost (AC):** The realized cost incurred for work performed on an activity during a given time period.

2. Performance Formulas:

```
Cost Variance (CV) = EV - AC Schedule Variance (SV) = EV - PV Cost Performance Index = CPI = EV / AC Schedule Performance Index = SPI = EV / PV Estimate at Completion = EAC = BAC / CPI Variance at Completion = VAC = BAC - EAC
```

3. Drill Tower Case Study Application (Status Month 8):

DRILL TOWER METRICS AT MONTH 8 (WEEK 36):
• Baseline Budget (BAC) = $26,500,000 | Planned Value (PV) = $20,500,000 (77.4% Planned)
• Earned Value (EV) = $17,540,000 (66.2% Complete) | Actual Cost (AC) = $23,440,000
• Cost Variance (CV) = $17.54M - $23.44M = -$5,900,000 (-33.6% Over budget)
• CPI Efficiency = $17.54M / $23.44M = **0.7483** (The project earns $0.75 of value for every $1.00 spent)
• Outturn Cost EAC = $26.50M / 0.7483 = **$35,413,604** (Predicted cost overrun: -$8,913,604).

Chapter 2: Advanced Earned Schedule (ES) & Time-Based Performance

A classic limitation of traditional EVM is that Schedule Variance (SV = EV - PV) is measured in currency, and as a project nears 100% completion, EV naturally converges to BAC, forcing SV -> 0 and SPI -> 1.0, even if the project is months late! Earned Schedule (ES), developed by Walt Lipke, resolves this by translating Earned Value into time units by identifying the exact calendar point where the current EV equaled planned baseline PV.

1. Earned Schedule Formulas:

```
Earned Schedule (ES) = C + (EV - PV_C) / (PV_C+1 - PV_C) Time Schedule Variance = SV_t = ES - Actual Months (AT) Time Performance Index = SPI_t = ES / AT Time Forecast Completion = EAC_t = Planned Duration (PD) / SPI_t
```

DRILL TOWER CASE STUDY APPLICATION:
• At Month 8 (AT = 8.0), Earned Value EV = $17.54M.
• Interpolating against baseline PV yields **Earned Schedule ES = 7.40 Months**.
• Time Schedule Variance SV_t = 7.40 - 8.00 = -0.60 Months = **-18.2 Days** delay.
• Time Velocity Index SPI_t = 7.40 / 8.00 = **0.9250**.
• Predicted Completion Date EAC_t = 12.0 / 0.9250 = **13.0 Months** (**January 31, 2027** / +31 days late).

Chapter 3: Quantitative Schedule & Cost Risk Analysis (Monte Carlo QCRA/QSRA)

Deterministic single-point estimates (EAC_1) fail to account for risk uncertainty and task correlation. Monte Carlo risk simulation executes 10,000 statistical iterations, sampling triangular risk distributions (Optimistic, Most Likely, Pessimistic) across all WBS control accounts to generate cumulative probability distribution curves (S-Curves).

1. Drill Tower Monte Carlo Percentile Results (10,000 Iterations):

Percentile	Confidence Level	Outturn Cost (EAC)	Cost Overrun vs BAC	Completion Date	Contingency Reserve
P10	10% Optimistic	$32,444,302	-$5,944,302	Feb 13, 2027	$0.00
P50	50% Median Outturn	$34,060,783	-$7,560,783	Feb 27, 2027	$0.00
P80	80% Standard Budget	$35,195,026	-$8,695,026	Mar 09, 2027	$0.00
P90	90% High Confidence	$35,815,202	-$9,315,202	Mar 14, 2027	+$401,598 Reserve
P95	95% Extreme Risk	$36,272,986	-$9,772,986	Mar 18, 2027	+$859,382 Reserve

P90 CONTINGENCY RESERVE RULE:
To achieve 90% confidence of completing within approved capital limits, project controllers must provision a total outturn budget of **$35,815,202**, requiring a management risk contingency reserve of **+$401,598** above the base EAC.

Chapter 4: Commercial Capital Budgeting & Investment Appraisal

Project Controllers must evaluate project cost overruns against commercial hurdle rates. Using Discounted Cash Flow (DCF) techniques under a 10.0% WACC discount rate, we evaluate commercial viability.

```
Net Present Value (NPV) = Sum [ CashFlow_t / (1 + r)^t ] - Outturn_CAPEX Internal Rate of Return = IRR where NPV = 0 Profitability Index = PI = PV of Inflows / Outturn_CAPEX
```

DRILL TOWER COMMERCIAL APPRAISAL RESULTS:
- Outturn CAPEX ($EAC) = $35,413,604 | Operating Cash Flow = $8.00M/year for 10 years
- **Net Present Value (NPV @ 10% WACC)** = **+$14,899,563** (Generates +$14.90M net wealth above hurdle)
- **Internal Rate of Return (IRR)** = **18.86%** (Exceeds 10.0% hurdle rate by +886 bps)
- **Simple Payback Period** = **4.43 Years** (53.1 Months / May 2031)
- **Profitability Index (PI)** = **1.42** (Generates $1.42 PV per $1.00 spent).

Chapter 5: Cash Burn Speed & Budget Depletion Runway Analysis

BUDGET DEPLETION RUNWAY FINDINGS:
- Average Monthly Burn Speed = **$2,930,000 / month** (vs planned $2.56M/month).
- Remaining Baseline Capital at Month 8 = **$3,060,000**.
- **100% BUDGET BURN OUT POINT:** The baseline budget ($BAC = $26.50M) will be completely exhausted in **Month 9 (September 2026)** ($AC_{\text{cum}} = \$26.64\text{M}$).
- **Executive Directive:** Secure C-suite approval for a **+$8.91M overrun credit line** before September 15, 2026.

Chapter 6: Variance Reconciliation & Cost Waterfall Bridges

A Cost Variance Waterfall Bridge reconciles the step-by-step progression from the original BAC budget to the final outturn EAC forecast, isolating scope overruns and time delay overhead extensions.

Step	WBS Code	Control Account / Driver	Incremental Cost	Cumulative Cost	% Share
0	1.0	Baseline Budget (BAC)	$26,500,000	$26,500,000	0.0%
1	1.1.1	Detail Structural Engineering Revisions	+$300,000	$26,800,000	3.4%
2	1.2.1	Procurement Steel Inflation	+$600,000	$27,400,000	6.7%
3	1.3.1	Verdal Sub-Structure Fabrication	+$200,000	$27,600,000	2.2%
4	1.3.2	Egersund Derrick Mast Yard Rework	+$2,400,000	$30,000,000	26.9%
5	1.4.1	Heavy Lift Vessel Weather Standby	+$1,300,000	$31,300,000	14.6%
6	1.4.2	Offshore Topside Lifting & Mating	+$100,000	$31,400,000	1.1%
7	1.5.1	Hook-up & Commissioning Crew	+$200,000	$31,600,000	2.2%
8	EAC_t	Time Delay Overhead Spread (SPI_t)	+$3,813,604	$35,413,604	42.8%
END	1.0	Final Outturn Forecast (EAC)	+$8,913,604	$35,413,604	100.0%

Chapter 7: Risk Heatmaps & Executive Governance Matrix

Risk heatmaps combine Probability (1-5) and Impact (1-5) to prioritize active risk exposure. Expected Monetary Value ($EMV = \text{Probability} \times \text{Financial Exposure}$) quantifies unmitigated risk reserves.

Risk ID	Risk Title	Category	Score	Exposure	EMV Value	Accountable CAM
R01	Egersund Derrick Mast Yard Rework	Fabrication	5x5 Critical	$2,400,000	$2,400,000	Lars Hansen
R02	North Sea Weather Standby Delay	Marine	4x4 High	$1,800,000	$720,000	Erik Solberg
R03	Tubular Steel Price Inflation	Procurement	3x3 Medium	$600,000	$180,000	Ingrid Berg
R04	DNV Structural Redesign Review	Engineering	2x3 Medium	$300,000	$60,000	Geir Nilsen
R05	Offshore Hook-up Crew Bottleneck	Commissioning	2x2 Low	$200,000	$40,000	Bjørn Lie

---

HANDBOOK SIGN-OFF & METHODOLOGICAL AUTHORSHIP:
Prepared by Frank Ellingsen, Financial Controller & Project Control Specialist.
Validated against EVM standards, Lipke Earned Schedule algorithms, and commercial WACC investment principles.